Seeing is believing

13a

Our eyes are one of our sense organs. They detect light.

On the outside, our eyes have protective **eyelids**. Closed, the eyelids protect eyes from smoke, onions, dust, bright lights and physical intrusion. The eyelids are surrounded by hairs called **eyelashes** and by touch receptors, that work together to protect our eyes.

Eyes are cleaned and lubricated by tears that are produced in the **tear gland**. Positioned near the eyebrows, the glands are not an external feature of the eye. Tears are released through **tear ducts** found near the corners of the eyes.

The coloured part of our eye is called the **iris**. It is a ring of muscle that changes the size of the **pupil**, the dark central area. Light passes through the pupil. If the light is too bright, the iris decreases the size of the pupil. If there is not much light, the iris increases the size of the pupil.

A cross-section of the human eye reveals a thin, clear skin, the **cornea**, that covers the front of the eye. Its properties allow light to pass through it. All the light that enters the eye is organised by the cornea and **lens**. The lens is made from a clear translucent material and is surrounded by a **lens muscle**. When the lens muscle relaxes, it becomes a large ring and the lens is made thin; when it contracts, it forms a small ring and the lens becomes thick. The cornea and lens focus the light rays onto the **retina**, which is located at the back of the eyeball. If a viewed object is close to the eye, the lens thickens so that the image of the object is focused and clear on the retina. For a distant object the eye lens is flattened. The retina has light receptors that are sensitive to the colours of light and to light brightness. The image of the object is upside down on the retina. The receptors in the retina send electrical impulses to the brain via the **optic nerve**, which is located outside the eyeball at the back of the eye. The brain interprets the impulses and 'turns the image the right way up'.

1 Complete the table. The place and function cells for the eyelid have been filled as an example.

Part	Place	Function
Eyelid	Outside, at the front	Protects the eye from things like dust, smoke, physical injury
Eyelashes		
Tear ducts		
Cornea		
Iris		
Pupil		
Lens		
Lens muscle		
Retina		
Optic nerve		

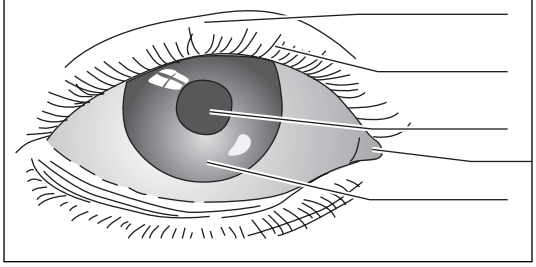
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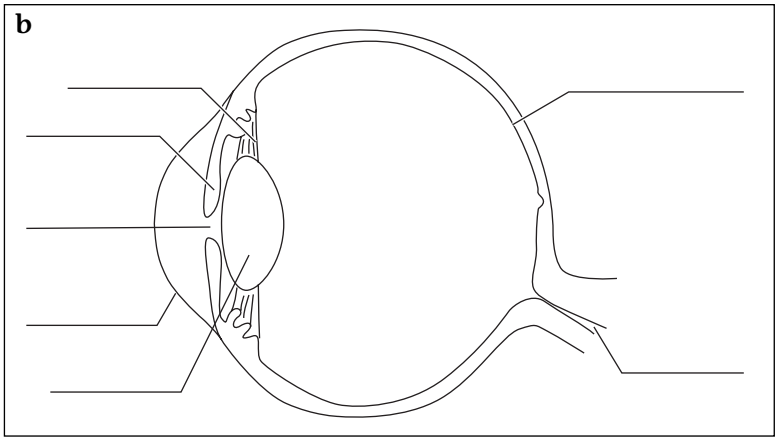
13b

2 Using words from the text above, label the arrowed features of each diagram.

a



b



3 Describe how the lens is adjusted when it focuses on an object that is:

a far away _____

b very close. _____

4 **a** What term could describe the type or style of text used at the beginning of this worksheet?

_____

b What is the purpose of this text type? _____

_____

c How many paragraphs does this text have? _____

d What does the first paragraph tell the audience? _____

e What does the rest of the text do? _____

_____

f A pronoun is a word used in place of a noun. We use pronouns (such as *they* and *it*) so we don't keep repeating the noun. Pronouns help to make the text clear to the reader. Underline three pronouns in the text about the eye, and draw an arrow to show which noun each underlined pronouns is replacing.

5 Use the text to help you unjumble the following sentences.

a an are detects Eyes important light. organ sense which

_____

b are by Eyes glands. in lubricated produced tears tear the

_____

c a clear cornea covers eye. front is of of part piece skin that the the The thin

_____

d image is on retina the The upside-down.

_____

e an and as eye focuses important incoming is it lens light. of organises part the the The

_____

_____

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